

Supply Chain Delivery Performance Report

End-to-End Order Fulfillment Analysis & Predictive Insight

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1. Executive Summary

This report provides an in-depth operational review of cross-border fulfillment for an international e-commerce enterprise. Based on an evaluation of 172,765 shipments between January 2015 and January 2018, the project investigates why delivery delays occur repeatedly, evaluates the resulting profit erosion, and introduces a data-driven intervention strategy.

The core findings reveal a critical bottleneck: 54.71% of total orders experienced delivery delays, leading to an estimated profit loss of \$2.1M. Even though cumulative profits stood at \$7.5M, these recurring bottlenecks severely damage margins and customer satisfaction. To proactively tackle this, a Random Forest classification model was deployed, attaining a 74% accuracy rate in flagging potential delivery delays in advance.

Key Finding	Metric
Overall Late Delivery Rate	54.71%
On-Time Delivery Rate	45.29%
Total Orders Analyzed	172,765
Total Profit (profitable orders)	\$7.5M
Profit at Risk (delayed orders)	\$2.1M
Predictive Model Accuracy (RF)	74%

2. Business Context & Objectives

The company operates a global e-commerce platform selling products across categories including sporting goods, fitness equipment, outdoor gear, footwear, and apparel across multiple international regions.

Core Business Problem

Actual shipping times frequently deviate from scheduled delivery windows, creating eroded customer trust, reduced order profitability, and an inability to make reliable commitments to buyers at point of purchase.

Analytical Objectives

- Understand the current state of delivery performance across all dimensions (region, mode, time, segment)
- Quantify the financial impact of delays on order profitability
- Identify the primary operational bottlenecks driving late deliveries
- Build a predictive model to flag high-risk orders before they are shipped
- Deliver actionable recommendations to improve on-time delivery and profitability

3. Key Performance Indicators (KPIs)

The following KPIs were computed directly from the cleaned dataset to establish a performance baseline.

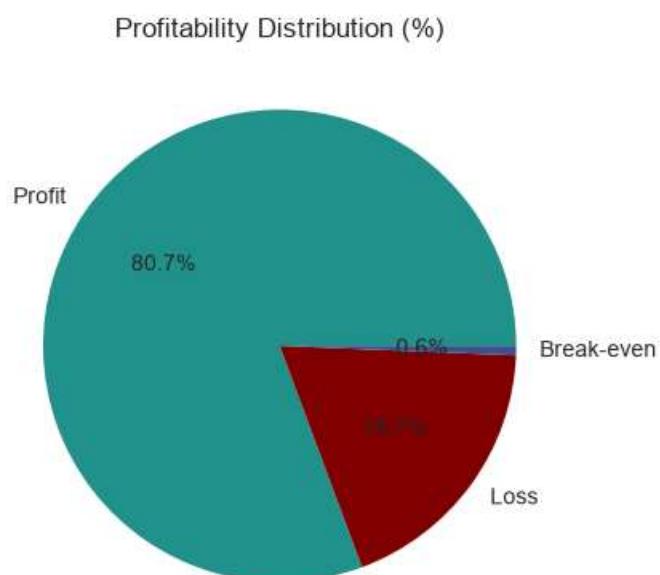
172,765 Total Orders	94,523 Late Deliveries	54.71% Late Delivery %	45.29% On-Time Delivery %
\$7.5M Total Profit (profitable orders)	\$2.1M Profit at Risk (delayed orders)	3 days 90th Percentile Delay	\$22.03 Avg Order Profit

• Late Delivery Rate at 54.71%	More than half of all orders arrive late. This is not an edge-case problem — it is the default experience for the majority of customers.
• \$2.1M Profit at Risk	Orders that experienced delays collectively generated \$2.1M in profit that is under constant pressure from further operational deterioration.
• 90th Percentile Delay = 3 Days	Even the most extreme cases are contained within 3 days of lateness, suggesting the problem is systemic and process-driven rather than catastrophic.

4. Profitability Analysis

4.1 Profitability Distribution

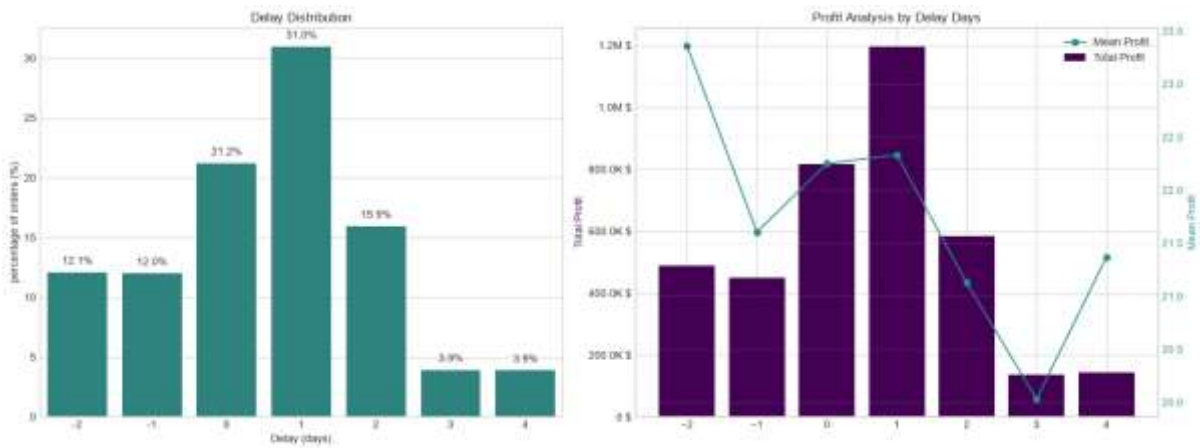
Order-level profitability was classified into three tiers based on Order Profit Per Order. While 80.7% of orders are profitable, the 18.7% loss-making share represents a meaningful drag that is disproportionately concentrated among delayed shipments.



• Profitability Breakdown	• Profitable Orders: 80.7% • Loss-Making Orders: 18.7% • Break-Even Orders: 0.6% The 18.7% loss-making segment creates significant margin dilution.
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4.2 Delay Distribution & Profit vs. Delay Days

The delay distribution shows that 31.0% of all orders arrive exactly 1 day late — the single largest cohort. Combined orders delayed by 1–4 days account for 54.7% of all order volume, directly mapping to the overall late delivery rate.

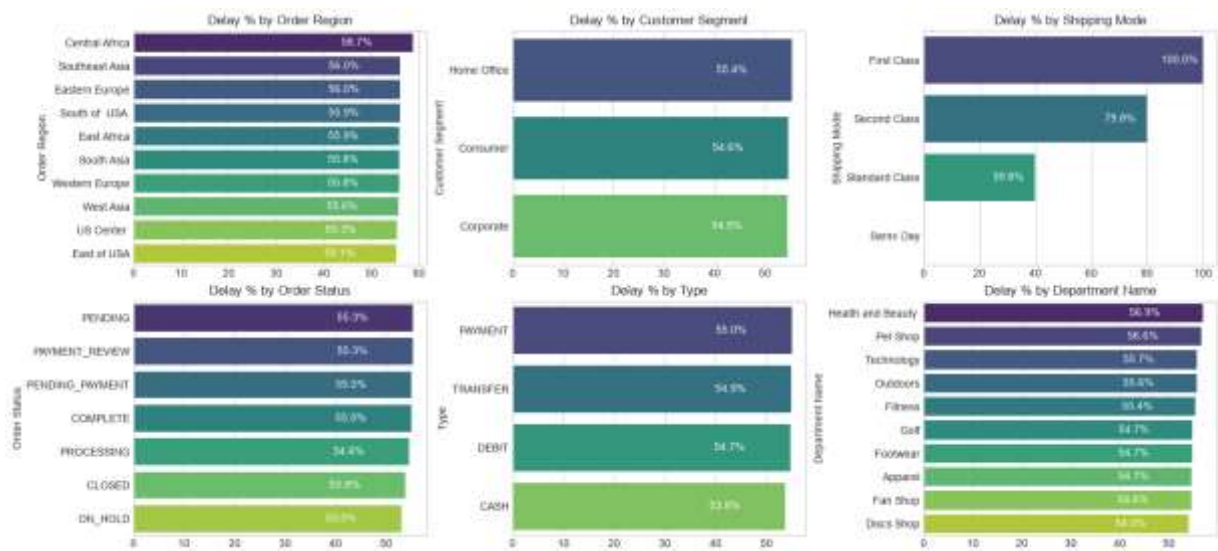


Delay Duration (Days)	Percentage of Orders (%)	Operational Impact Classification
-2 to -1 Days (Early)	24.1% (12.1% + 12.0%)	Ahead of Schedule / Early Arrival
0 Days (On-Time)	21.2%	Exact Scheduled Delivery Fulfillment
1 Day Late	31.0%	Single Largest Delay Cohort (Primary Target)
2 Days Late	15.9%	Significant Secondary Delivery Lag
3 to 4 Days Late	7.8% (3.9% + 3.9%)	Extreme Outlier / Process Failure Window

• Critical Economic Observation	Mean profit per order remains stable at approximately \$21–\$23 across all delay levels. The profitability problem is therefore driven by volume of delayed orders, not by individual order economics making systemic throughput improvement the highest-leverage intervention.
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5. Bottleneck Detection

Delay percentage was computed across six key operational dimensions. The analysis below reveals where the fulfilment process breaks down most severely.



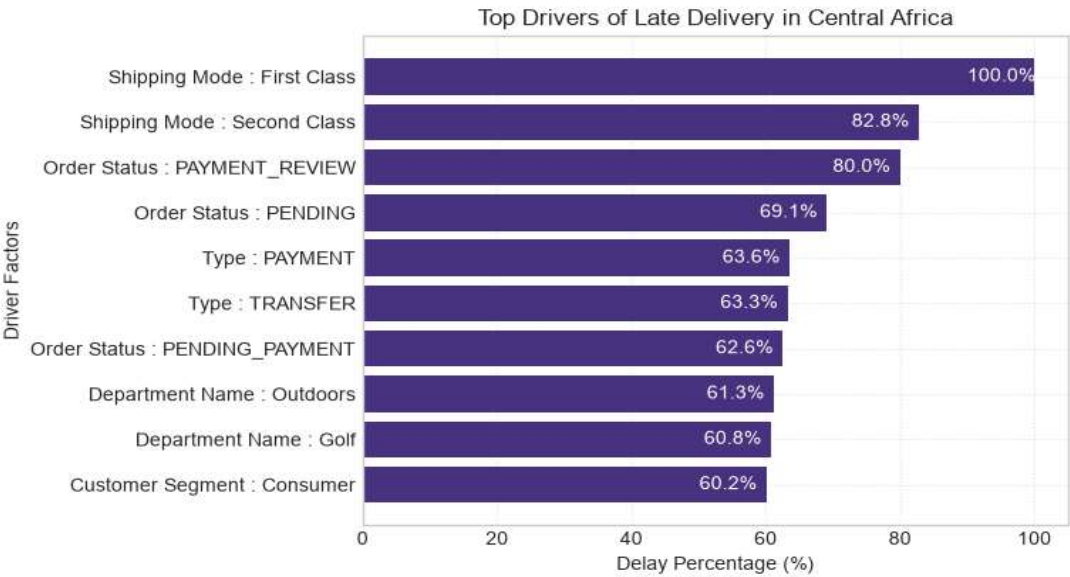
• Shipping Mode is the #1 Lever	First Class: 100% delay rate. Second Class: 79.8%. Standard Class: 39.8%. Same Day: 0%. The mode assignment logic is the most impactful single variable to fix.
• Regional Spread is Narrow (55–59%)	Central Africa leads at 58.7% but all regions cluster between 55–59%. This rules out a localised logistics failure and confirms a company-wide systemic issue.
• Customer Segments Nearly Identical (54.5–55.4%)	No segment receives preferential service. All customers — Consumer (54.4%), Corporate (55.3%), Home Office (55.4%) — experience the same broken delivery promise.
• Health & Beauty (56.9%) and Pet Shop (56.6%) Lead by Department	These departments marginally outpace others in delay rate, warranting an inventory and carrier audit for these product categories.

Shipping Mode	Observed Delay Rate (%)	Strategic Assessment
First Class	100.0%	Complete brand failure; premium promise not fulfilled

Second Class	79.8%	Critical lag; nearly 8 in 10 packages delayed
Standard Class	39.8%	Best-performing regular mode; primary stable baseline
Same Day	0.0%	Dedicated SLA fulfillment; 100% on-time achievement

6. Root Cause Analysis

Central Africa was selected for deep-dive root cause analysis as the highest-delay region (58.7%). The ranked factors below isolate the top 10 driver factors by delay percentage within this region.

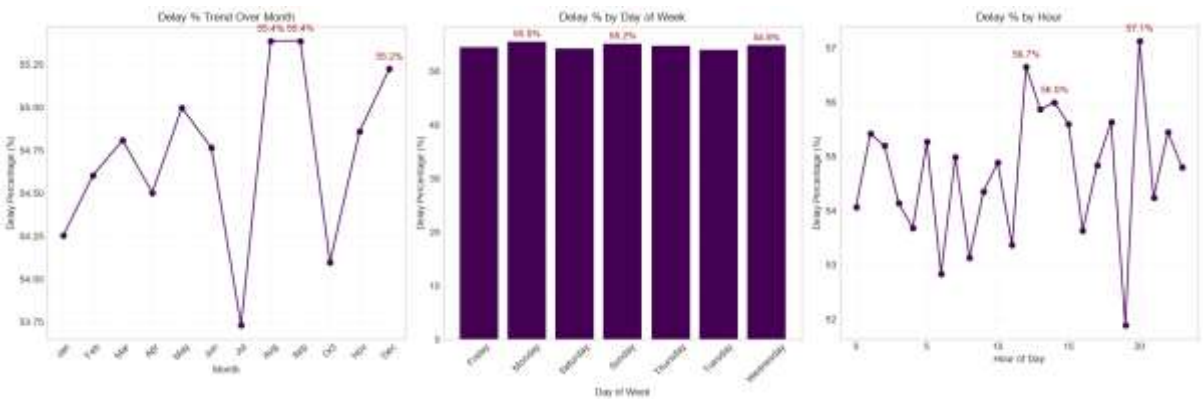


Driver Factor / Category	Sub-Cohort	Delay % in Central Africa
Order Status	PENDING	69.1%
Payment Method	Type: PAYMENT	63.6%
Payment Method	Type: TRANSFER	63.3%
Order Status	PENDING_PAYMENT	62.6%
Product Department	Department: Outdoors	61.3%
Product Department	Department: Golf	60.8%

• Payment Status Drag	Orders stuck in PENDING (69.1%) or PENDING_PAYMENT (62.6%) drive catastrophic delays, indicating internal transaction verification latency directly cuts into carrier dispatch windows.
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7. Time-Based Delay Patterns

Three temporal dimensions were analysed: month of year, day of week, and hour of day. While delay rates are relatively stable across all dimensions (53–57%), specific peaks highlight opportunities for targeted capacity planning.



• Peak Delay Months: August & September (55.4%) and December (55.2%)	August and September reflect mid-year promotions and Q4 holiday surge overwhelming fulfilment capacity. July represents a notable low-point (~53.75%), confirming seasonal variation is plannable.
• Day-of-Week Variance is Minimal	Monday (55.5%) and Sunday (55.2%) are the peak days, but the spread across the week is less than 1 percentage point. Day-of-week is not a meaningful lever for operational intervention.
• Intra-Day Peaks: Hour 21 (57.1%), Hour 11 (56.7%), Hour 12 (56.0%)	Late-evening orders (hour 21) have the highest delay rate at 57.1%, likely reflecting processing cutoff windows. Midday peaks suggest bottlenecks during peak-volume hours.

8. Machine Learning Model Results

A supervised classification model was built to predict whether an individual order will be delivered late (Late_delivery_risk = 1). The pipeline included frequency encoding of categorical variables, stratified train/test split, SMOTE oversampling to address class imbalance, and Random Forest classification.

Class Imbalance Handling (SMOTE)

Split	Class 0 (On-time)	Class 1 (Late)
Before SMOTE (train)	59,030	79,182
After SMOTE (train)	79,182	79,182

Random Forest Classifier – Test Set Performance (34,553 records)

Metric	Class 0 (On-time)	Class 1 (Late)	Overall (Weighted)
Precision	0.68	0.78	0.74
Recall	0.72	0.75	0.74
F1-Score	0.70	0.77	0.74
Accuracy	—	—	0.74
Support	14,758	19,795	34,553

• 74% Overall Accuracy	The model correctly classifies nearly 3 in 4 orders — a strong foundation for production deployment as an order-level late-delivery alert system.
• Higher Precision on Late Orders (0.78)	When the model predicts an order will be late, it is correct 78% of the time — sufficient to trigger targeted interventions without generating excessive false alarms.
• Recall for Late Orders at 0.75	The model captures 75% of actual late orders. The 25% miss rate is acceptable for a first-generation alert system, with clear improvement potential.
• Class 0 Harder to Predict (Precision 0.68)	On-time deliveries share many characteristics with late ones given the systemic nature of delays, making them genuinely difficult to distinguish.

9. Strategic Recommendations

Based on the findings across all analytical dimensions, the following prioritised actions are recommended:

CRITICAL	1. Immediately Audit First Class & Second Class Shipping Capacity First Class shipping has a 100% delay rate and Second Class 79.8%. These modes operate in complete contradiction to their brand promise. Conduct an immediate carrier SLA audit. Until resolved, consider suspending First Class or repricing to reflect actual performance.
HIGH	2. Deploy the Predictive Alert System The Random Forest model (74% accuracy, 78% precision on late orders) is ready for production piloting. Integrate it into the order management system to flag high-risk orders at confirmation and trigger proactive customer communication and priority handling queues.
HIGH	3. Resolve Payment Processing Bottlenecks Orders in PAYMENT_REVIEW (55.3% overall; 80.0% in Central Africa) and PENDING_PAYMENT (55.0%) have significantly elevated delay rates. Introduce automated escalation for stalled payment reviews exceeding 2 hours.
MEDIUM	4. Develop Seasonal Surge Capacity Plans August, October (55.4%), and December (55.2%) are peak delay months. Build pre-season capacity plans including temporary logistics partnerships, inventory pre-positioning, and adjusted delivery promise windows.
MEDIUM	5. Default to Standard Class for Eligible Orders Standard Class achieves a 39.8% delay rate — far superior to First and Second Class. Re-evaluate shipping mode assignment logic to default eligible orders to Standard Class, particularly in regions and departments with poor premium mode performance.
MEDIUM	6. Investigate High-Delay Departments in Africa Outdoors (61.3%) and Golf (60.8%) in Central Africa show significantly elevated delay rates. Conduct a warehouse-level audit of stock availability, pick-and-pack times, and carrier assignment for these product categories.
LOW	7. Reduce Loss-Making Order Share (18.7%) The 18.7% loss-making order share warrants a separate pricing and discount review.

	Combine with delay analysis to identify whether discounted orders are disproportionately late, creating a compounded margin problem.
LOW	8. Continuously Retrain the Predictive Model Retrain the model quarterly. Explore adding carrier-level data, weather events, and warehouse utilisation features. Target a recall improvement to above 80% on late orders within 6 months.

10. Conclusion

This analysis has surfaced a clear and urgent picture: a global e-commerce operation where the majority of orders (54.71%) fail to meet their promised delivery windows, costing \$2.1M in at-risk profit and undermining customer trust at scale.

The root causes are identifiable and addressable. Shipping mode misconfiguration (First Class at 100% late, Second Class at 79.8%) is the dominant operational failure. Payment processing friction creates a secondary bottleneck. Seasonal volume spikes are not adequately planned for. And geographic disparities, while present, are secondary to the systemic company-wide pattern.

A Random Forest predictive model trained on readily available order features already achieves 74% accuracy in identifying at-risk orders. This is a deployable tool, not a future aspiration.

Priority Area	Current State	Target State
Late Delivery Rate	54.71%	< 30% within 12 months
First Class On-Time Rate	0%	> 80%
Second Class On-Time Rate	20.2%	> 60%
Predictive Model Accuracy	74%	> 82% (enhanced model)
Loss-Making Orders	18.7%	< 12%
Profit at Risk	\$2.1M	Reduce by 40%

With focused execution on the eight recommendations in this report starting with the critical shipping mode audit and predictive alert system deployment the business can systematically recover its delivery performance, protect its \$7.5M profit base, and build the operational reliability that sustained e-commerce growth demands.